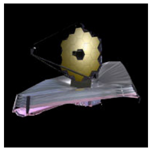

7985 - Resolving the thin disk of a nearby Milky Way analog

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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Dr. Tony Wong (CoI)	University of Illinois at Urbana - Champaign

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	MIRI mosaic of NGC 4565	MIRI Imaging	(1) NGC-4565
	2	NIRCam mosaic of NGC 4565	NIRCam Imaging	(1) NGC-4565

ABSTRACT

We propose panoramic MIRI and NIRCam mapping of the prototype nearby edge-on galaxy, NGC 4565 (inclination = 87.5 degrees). With < 20h total time, we will resolve the vertical structure of PAH emission, dust continuum, and multiple star formation tracers over the whole disk. Measurements of vertical structure are critical to test models of self-regulation in galaxies. Such vertical self-regulation is predicted by simulations and widely used to interpret observations of all types of galaxies but not yet strongly tested by direct, edge-on observations of other galaxies. JWST imaging will change that, achieving < 20 pc resolution and detecting emission all the way out to galactocentric radius > 30 kpc, far into the outer, HI-dominated regime. These observations are also ideal to identify and characterize giant interstellar filaments, which appear as extended in-plane structures in Milky Way dust observations. Thus, this program will test the generality of an emerging new paradigm in ISM structure.

OBSERVING DESCRIPTION

We design MIRI and NIRCam mosaics that cover the disk out to where Spitzer imaging indicates the 8 micron imaging to drop below 0.5 MJy/sr. We obtain the NIRCam observations in parallel with the required MIRI background observations. The background regions appear free of mid-IR emission in previous imaging, and they cover a wide area to easily suppress any unexpected artifacts. The long, thin disk requires that we moderately constrain the position angle and then we use row offsets to align the mosaic with the disk.

We adopt readout and dithering patterns that have worked successfully for previous mosaic imaging of nearby galaxies (GO 2107, GO 3707). We also match those integration times, consistent with the observed brightness and outer brightness contour for our sources.

Proposal 7985 - Targets - Resolving the thin disk of a nearby Milky Way analog

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4565	RA: 12 36 20.8040 (189.0866833d)	Epoch of Position: 2000	
		Dec: +25 59 14.61 (25.98739d)		
		Equinox: J2000		

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Spiral galaxies]

Observation

Proposal 7985, Observation 1: MIRI mosaic of NGC 4565

Tue Apr 22 17:00:57 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:13) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:14) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:15) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:16) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:17) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:18) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:19) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:20) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4565	RA: 12 36 20.8040 (189.0866833d) Dec: +25 59 14.61 (25.98739d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Spiral galaxies]

Template

Subarray
FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
10	2	15.0	10.0	9.0	0.0	DEFAULT

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4						LARGE

Proposal 7985 - Observation 1 - Resolving the thin disk of a nearby Milky Way analog

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F770W	FASTR1	8	1	1	Dither 1	4	4	88.801	227297
2	F2100W	FASTR1	15	2	1	Dither 1	4	8	344.105	227297

Special Requirements

Group Visits within 53.0 Days
Aperture PA Range 123.83544897 to 129.83544897 Degrees (V3 119.0 to 125.0)
Visits Same PA

Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 7985, Observation 2: NIRCam mosaic of NGC 4565 Tue Apr 22 17:00:57 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Coordinated Parallel Template(s): MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4565	RA: 12 36 20.8040 (189.0866833d) Dec: +25 59 14.61 (25.98739d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Spiral galaxies]

Template

NIRCam Imaging	MIRI Imaging
Module: B Subarray: FULL	Subarray: FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
8	1	15.0	10.0	13.5	0.0	DEFAULT

Dithers

#	Primary Dither Type	Primary Dithers	Dither Size	Subpixel Positions	Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
1	INTRAMODULEBOX	4		1	NIRCam Only	NO_DITHERING

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F187N	F335M	BRIGHT2	5	1	4	4	429.471	227297
2	F200W	F300M	BRIGHT2	3	1	4	4	257.682	227297

Proposal 7985 - Observation 2 - Resolving the thin disk of a nearby Milky Way analog

Spectral Elements

MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F2100W	FASTR1	15	2	1		4	8	344.105	
2	F770W	FASTR1	8	1	1		4	4	88.801	

Special Requirements

Sequence Visits within 53.0 Days
Aperture PA Range 120.05262691 to 125.05262691 Degrees (V3 120.0 to 125.0)
Visits Same PA
No Parallel Attachments

Sequence Observations 1, 2, Non-interruptible